

Problem 1: That's a lot of friends!

Problem 1. *Sehtaj has 10001 friends. Some of them join together to form several groups (a friend may belong to different groups). Some groups join together to form several societies (a group may belong to different societies). There are a total of k societies. Suppose that the following conditions hold:*

- (i) *Each pair of friends is in exactly one group.*
- (ii) *For each friend and each society, the friend is in exactly one group of the society.*
- (iii) *Each group has an odd number of friends. In addition, a group with $2m + 1$ friends (m is a positive integer) is in exactly m societies.*

Find all possible values of k .

Answer

Answer: $k = 5000$.

Solution: In order to use all the information in the question, we count triples (a, C, S) , where a is a friend, C is a group and S is a society, where $a \in C$ and $C \in S$. Let the number of such triples be T .

Suppose we first fix a , then S , then C . We can choose a in 10001 ways, S in k ways and then finally C in only one way (by condition (ii)). Hence $T = 10001 \cdot k$.

Now suppose we fix C first. There are $|C|$ ways of doing this. Then by condition (iii), there are $(|C| - 1)/2$ ways to choose S . Finally there is only one way to choose a , by (ii). This gives

$$T = \sum_{C \text{ is a group}} |C|(|C| - 1)/2 = \sum_{C \text{ is a group}} \binom{|C|}{2}$$

On the other hand, the sum $\sum_{C \text{ is a group}} \binom{|C|}{2}$ is actually equal to the number of pairs of friends. This is because each pair of friends is in exactly one group by (i), so each pair of friends is counted exactly once. Hence this sum is equal to $\binom{10001}{2}$, so putting everything together

$$\binom{10001}{2} = T = 10001 \cdot k \implies k = 5000.$$

Finally, to construct a configuration for $k = 5000$, let there be only one group (C) containing all 10001 friends, and let this group be in all 5000 societies. Since $|C| = 10001 = 2(5000) + 1$, by condition (iii), this group should be in exactly 5000 societies, which is satisfied. Condition (i) is satisfied since all pairs of friends are in the one group C . Condition (ii) is satisfied since for each friend and each society, the friend is in exactly one group (namely C) of that society. ■

proposed by Tawhid Bin Omar

Problem 2: Finding the Heaviest Coin

Problem 2. *We are given n coins of different weights and n balances, $n > 2$. On each turn one can choose one balance, put one coin on the right pan and one on the left pan, and then remove these coins from the balance. It is known that one balance is wrong (but it is not known which exactly), and it shows an arbitrary result on every turn. What is the smallest number of turns required to find the heaviest coin?*

Answer

Answer: $2n - 1$.

Solution: Let $c_1, c_2, \dots, c_n$ be the coins. Let b_1, b_2, b_3 be three balances.

Construction: We use the following strategy: Compare coins c_i and c_j using balances b_1 and b_2 . If both balances agree that $c_i > c_j$ (or both agree that $c_j > c_i$), then we trust this result and continue. We then compare c_i (or whichever coin was heavier) with a new coin c_k using the same two balances b_1 and b_2 .

If at some point balances b_1 and b_2 give different results (one says $c_i > c_j$ and the other says $c_j > c_i$), then we know that one of these two balances is the faulty one. For all remaining comparisons, we use balance b_3 (which we now know is reliable) along with one of b_1 or b_2 .

Let x be the number of comparisons we make before finding a disagreement. Then:

- We make $2x$ weighings to find the disagreement (using b_1 and b_2)
- We make 2 more weighings with the disagreement itself
- We make $(n - x - 1)$ more weighings using b_3 and one other balance

Total number of weighings: $2x + 2 + (n - x - 1) = n + x + 1$.

In the worst case, $x = n - 2$ (we don't find a disagreement until we've compared all but one coin), giving us $n + (n - 2) + 1 = 2n - 1$ weighings. ■

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Problem 3: A Number Writing Game

Problem 3. *Onyx and Sehtaj play a game. Given an integer N , Onyx writes down 1 first, then every player sees the last number written and if it is n then in her turn she writes $n + 1$ or $2n$, but her number cannot be bigger than N . The player who writes N wins. For which values of N does Sehtaj win?*

Answer

Answer: Sehtaj wins if and only if $N = 2^k$ and k is odd.

Solution: If N is odd, Onyx wins. She first writes 1, an odd number. If she writes an odd number n , both $n + 1$ and $2n$ are even, so the number she gets is even and thus she can make the $+1$ move, writing another odd number. Thus Sehtaj only writes even numbers and never N .

Let $N = d_1 d_2 \cdots d_k$ in binary, and suppose N is even. Let $a_i = d_1 d_2 \cdots d_{k-i+1} + 1$. Partition the set $\{1, 2, \dots, N\}$ into $S_1 = \{N, N - 1, \dots, a_2\}$, $S_2 = \{a_2 - 1, \dots, a_3\}$, and so on. A move $n \rightarrow 2n$ always goes to the following subset. Now it is clear that if a player is the first to write a number in S_1 , doing it by doubling the previous number, she wins (because she writes the even numbers from then on). If X is the first to write a number in S_2 , Y will then jump to S_1 by an $n \rightarrow 2n$ move and win. So the first to write a number in S_2 loses. We have $S_2 = \{d_1 \cdots d_{k-1}, \dots, d_1 \cdots d_{k-2} + 1\}$. Clearly $d_1 \cdots d_{k-2}$ will be written, unless a player commits suicide, because the only way to skip it is to play an $n \rightarrow 2n$ move and that is losing. So for even N the player who writes $d_1 \cdots d_{k-2}$ wins.

Then the winner for the $d_1 \cdots d_k$ game is Onyx if $d_k = 1$, and it is the same as for the $d_1 \cdots d_{k-2}$ game if $d_k = 0$. If among d_k, d_{k-2}, d_{k-4} and so on there's a 1, the winner is Onyx. If not, and if k is even, the winner is the same as for $d_1 d_2$, which is Onyx if $d_2 = 1$, and Sehtaj if $d_2 = 0$, by inspection. If k is odd, the winner is either Onyx or the same as for $d_1 d_2 d_3$, which is always Onyx by inspection.

We conclude that Sehtaj wins if and only if N is the sum of distinct odd powers of 2. In particular, Sehtaj wins if and only if $N = 2^k$ and k is odd. ■

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Problem 4: Counting Special Permutations

Problem 4. Find the number of permutations $(p_1, p_2, p_3, p_4, p_5, p_6)$ of $1, 2, 3, 4, 5, 6$ such that for any k , $1 \leq k \leq 5$, $(p_1, \dots, p_k)$ does not form a permutation of $1, 2, \dots, k$.

Answer

Answer: 461.

Solution: Let's denote the set of permutations of $\{1, \dots, n\}$ as P_n , where $|P_n| = n!$ is the number of permutations. For a permutation $P = (\alpha_1 \dots \alpha_n)$, let's define the *degree of the permutation*, denoted as $\deg P$. (Note: The degree of a permutation can be a different concept in the literature; here we use our own definition.) For $1 \leq k \leq n$, if $(\alpha_1 \dots \alpha_k)$ represents a permutation of the set $\{1, \dots, k\}$, the degree is the smallest k for which this holds.

Let Q_n denote the set of permutations of $\{1, \dots, n\}$ such that, for every $1 \leq k \leq n-1$, $(\alpha_1 \dots \alpha_k)$ is not a permutation of $\{1, \dots, k\}$. For example, $Q_1 = \{(1)\}$ and $Q_2 = \{(21)\}$. The problem asks for $|Q_6|$.

For $1 \leq i \leq n$, let $D_i = \{P \in P_n \mid \deg P = i\}$. Note that D_i and D_j are disjoint for $i \neq j$, and $P_n = D_1 \cup \dots \cup D_n$. First, observe that $D_n = \{P \in P_n \mid \deg P = n\} = Q_n$.

If we append a permutation of $\{i+1, \dots, n\}$ to Q_i , we obtain a permutation in D_i . Therefore, we have the equation:

$$n! = |P_n| = |Q_n| + |Q_{n-1}| \cdot 1! + |Q_{n-2}| \cdot 2! + \dots + |Q_1| \cdot (n-1)! = \sum_{i=1}^n |Q_i| \cdot (n-i)!$$

Let's calculate for smaller values:

$$\begin{aligned} |P_3| = 3! &= |Q_3| + |Q_2| \cdot 1! + |Q_1| \cdot 2! \\ \implies |Q_3| &= 3! - 1 - 2 = 3. \end{aligned}$$

$$\begin{aligned} |P_4| = 4! &= |Q_4| + |Q_3| \cdot 1! + |Q_2| \cdot 2! + |Q_1| \cdot 3! \\ \implies |Q_4| &= 4! - 3 - 2 - 6 = 13. \end{aligned}$$

$$\begin{aligned} |P_5| = 5! &= |Q_5| + |Q_4| \cdot 1! + |Q_3| \cdot 2! + |Q_2| \cdot 3! + |Q_1| \cdot 4! \\ \implies |Q_5| &= 5! - 13 - 6 - 6 - 24 = 71. \end{aligned}$$

$$\begin{aligned} |P_6| = 6! &= |Q_6| + |Q_5| \cdot 1! + |Q_4| \cdot 2! + |Q_3| \cdot 3! + |Q_2| \cdot 4! + |Q_1| \cdot 5! \\ \implies |Q_6| &= 6! - 71 - 26 - 18 - 24 - 120 = 461. \quad \blacksquare \end{aligned}$$

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Problem 5: A Trigonometric Equation

Problem 5. Solve over $\mathbb{R}$ the equation

$$4^{(\sin x)^2} + 3^{(\tan x)^2} = 4^{(\cos x)^2} + 3^{(\cot x)^2}.$$

Answer

Answer: $x = \frac{\pi}{4} + k\frac{\pi}{2}$ for $k \in \mathbb{Z}$.

Solution: Setting $t = \cos^2 x \in (0, 1)$, we have:

$$\begin{aligned}\sin^2 x &= 1 - t, \\ \tan^2 x &= \frac{\sin^2 x}{\cos^2 x} = \frac{1 - t}{t}, \\ \cot^2 x &= \frac{\cos^2 x}{\sin^2 x} = \frac{t}{1 - t}.\end{aligned}$$

Substituting these into the equation, we obtain:

$$4^{1-t} + 3^{\frac{1-t}{t}} = 4^t + 3^{\frac{t}{1-t}}.$$

Let $f(t) = 4^t + 3^{\frac{t}{1-t}}$ and $g(t) = 4^{1-t} + 3^{\frac{1-t}{t}}$. We need to solve $g(t) = f(t)$.

Observe that:

- The function $f(t) = 4^t + 3^{\frac{t}{1-t}}$ is increasing on $(0, 1)$, since both 4^t and $3^{\frac{t}{1-t}}$ are increasing.
- The function $g(t) = 4^{1-t} + 3^{\frac{1-t}{t}}$ is decreasing on $(0, 1)$, since 4^{1-t} is decreasing and $3^{\frac{1-t}{t}}$ is decreasing.

Since $f(t)$ is increasing and $g(t)$ is decreasing, they can intersect at most once.

We can verify that $t = \frac{1}{2}$ is a solution:

$$g\left(\frac{1}{2}\right) = 4^{1/2} + 3^1 = 2 + 3 = 5, \quad f\left(\frac{1}{2}\right) = 4^{1/2} + 3^1 = 2 + 3 = 5.$$

Therefore, $t = \frac{1}{2}$ is the unique solution, which gives $\cos^2 x = \frac{1}{2}$.

This implies $\cos x = \pm \frac{1}{\sqrt{2}} = \pm \frac{\sqrt{2}}{2}$, so

$$\boxed{x = \frac{\pi}{4} + k\frac{\pi}{2} \text{ for } k \in \mathbb{Z}}. \quad \blacksquare$$

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Problem 6: A Board Game Strategy

Problem 6. *Sehtaj and Onyx play alternately on a 5×5 board. Sehtaj always enters a 1 into an empty square, and Onyx always enters a 0 into an empty square. When the board is full, the sum of the numbers in each of the nine 3×3 squares is calculated and Sehtaj's score S is the largest such sum. What is the largest score Sehtaj can make, regardless of the responses of Onyx?*

Answer

Answer: 6.

Solution: Tile the board with 10 dominoes (marked D in the figure below), leaving the bottom row untiled.

D	D	D	D	D
D	D	D	D	D
D	D	D	D	D
D	D	D	D	D

Onyx can ensure that each domino contains at least one 0, since whenever Sehtaj plays in a domino, Onyx writes a 0 in the other square. Each 3×3 square has at least 3 full dominoes, and hence will have at least three 0s. Thus Onyx can ensure that S is at most 6.

Now we show that Sehtaj can ensure that S is at least 6. Sehtaj can focus on filling the top-left 3×3 square with as many 1s as possible. Since the top-left 3×3 square contains exactly 3 dominoes (covering 6 squares) and 3 additional squares from the bottom row of this 3×3 region, Sehtaj can guarantee at least 6 ones in this square by playing strategically. Therefore, Sehtaj can ensure that $S \geq 6$.

We conclude that the largest score Sehtaj can make is 6. ■

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